

Where to act, and what to build

Two pillars. **Pillar 1 — Prediction:** models on existing UK open data to target help, with **mental health and ageing as the primary domains**. **Pillar 2 — 4th Space:** a physical/social venue (sauna, cold plunge, contrast therapy) that combats loneliness by bringing people together. This pack models both.

1 — Composite risk, tier and 3-period trend

#	PLACE	RISK	TIER	PROJECTED H1/H2/H3 (±BAND)
1	Blackpool	95.2	Critical	94.7/94.2/93.7 (±6.5)
2	Middlesbrough	93.6	Critical	96.1/98.6/100 (±6.5)
3	Glasgow	90.6	Critical	90.1/89.6/89.1 (±6.5)
4	London (Newham)	88.7	Critical	91.2/93.7/96.2 (±6.5)
5	Manchester	88.1	Critical	90.6/93.1/95.6 (±6.5)
6	Kingston upon Hull	87.6	Critical	90.1/92.6/95.1 (±6.5)
7	Nottingham	85.6	Critical	88.1/90.6/93.1 (±6.5)
8	UK (national)	85.0	Critical	87.5/90.0/92.5 (±6.5)
9	England (national)	84.7	Critical	87.2/89.7/92.2 (±6.5)
10	Liverpool	83.6	Critical	86.1/88.6/91.1 (±6.5)

2 — Pillar 2: 4th Space model (flagship loneliness intervention)

4th Space — sauna, cold plunge, contrast therapy

A physical/social venue that combats loneliness by bringing people together through shared contrast-therapy rituals. Models the CIC's pillar-2 delivery.

Members/site: **60** · Expected loneliness reduced: **48** · SROI proxy: **x3.2**

Est. annual investment: **£4,320** → est. social value: **£13,824**

- Sauna + cold plunge membership (subsidised, social sessions) (gain 0.80, ~£540)
- Guided contrast-therapy + community circle (gain 0.78, ~£480)
- Intergenerational 4th-space sessions (ageing + youth) (gain 0.72, ~£520)

3 — Budget optimisation (£120,000)

£54,960.0 21 ~428.0 £65,040.0 DEPLOYED PLACES COVERED EST. PEOPLE HELPED UNALLOCATED				
PLACE	TIER	INTERVENTION	COST	~PEOPLE

Blackpool	Critical	Community navigator + befriending	£480	~7.8
Glasgow	Critical	Community navigator + befriending	£480	~7.8
Blackpool	Critical	Social prescribing + group activity	£460	~7.2
Glasgow	Critical	Social prescribing + group activity	£460	~7.2
Blackpool	Critical	Transport-to-service vouchers	£390	~5.2
Glasgow	Critical	Transport-to-service vouchers	£390	~5.2
Middlesbrough	Critical	Debt advice + money guidance (CHW-delivered)	£650	~6.3
Manchester	Critical	Debt advice + money guidance (CHW-delivered)	£650	~6.3
Kingston upon Hull	Critical	Debt advice + money guidance (CHW-delivered)	£650	~6.3
Nottingham	Critical	Debt advice + money guidance (CHW-delivered)	£650	~6.3
Liverpool	Critical	Debt advice + money guidance (CHW-delivered)	£650	~6.3
Middlesbrough	Critical	Targeted child-poverty cash / cost-of-living support	£1,100	~8.1
Manchester	Critical	Targeted child-poverty cash / cost-of-living support	£1,100	~8.1
Kingston upon Hull	Critical	Targeted child-poverty cash / cost-of-living support	£1,100	~8.1
Nottingham	Critical	Targeted child-poverty cash / cost-of-living support	£1,100	~8.1
Liverpool	Critical	Targeted child-poverty cash / cost-of-living support	£1,100	~8.1
+45 more				

4 — Causal graph: lever to expected outcome

PRESSURE	LEVER	EXPECTED OUTCOME
mental	vcse_talking_therapy_nav	access_to_treatment
mental	low_intensity_cbt	reduced_symptoms
mental	peer_support	reduced_loneliness_crisis
ageing	ageing_connector	reduced_older_isolation
ageing	fall_prevention_club	reduced_falls_hospitalisation
ageing	warm_space	reduced_fuel_poverty
isolation	4th_space_contrast	reduced_loneliness_connection
isolation	navigator_befriend	reduced_loneliness
isolation	group_activity	increased_belonging
poverty	cash_support	reduced_child_poverty
poverty	debt_advice	reduced_arrears_escalation
homelessness	tenancy_sustainment	prevented_lose_home
+3 more edges		

5 — Intervention systems (top-tier places)

Critical Blackpool risk 95.2

pressures: isolation

- **Community navigator + befriending** — gain 0.60, ~£480
- **Social prescribing + group activity** — gain 0.55, ~£460
- **Transport-to-service vouchers** — gain 0.40, ~£390

Stop / escalate: Review at week 8; stop expansion if completion <35% or access gap widens in Blackpool. Escalate to safeguarding lead if self-harm risk indicated.

Critical Middlesbrough risk 93.6

pressures: poverty

- **Targeted child-poverty cash / cost-of-living support** — gain 0.90, ~£1,100
- **Debt advice + money guidance (CHW-delivered)** — gain 0.70, ~£650
- **Living-wage / employment access programme** — gain 0.50, ~£1,400

Stop / escalate: Review at week 8; stop expansion if completion <35% or access gap widens in Middlesbrough. Escalate to safeguarding lead if self-harm risk indicated.

Critical Glasgow risk 90.6

pressures: isolation

- **Community navigator + befriending** — gain 0.60, ~£480
- **Social prescribing + group activity** — gain 0.55, ~£460
- **Transport-to-service vouchers** — gain 0.40, ~£390

Stop / escalate: Review at week 8; stop expansion if completion <35% or access gap widens in Glasgow. Escalate to safeguarding lead if self-harm risk indicated.

Critical London (Newham) risk 88.7

pressures: homelessness

- **Preventive tenancy sustainment (pre-Section-21 reach)** — gain 0.85, ~£950
- **Rapid rehousing + TA diversion** — gain 0.80, ~£1,300
- **Rent arrears mediation with landlord** — gain 0.60, ~£700

Stop / escalate: Review at week 8; stop expansion if completion <35% or access gap widens in London (Newham). Escalate to safeguarding lead if self-harm risk indicated.

Critical **Manchester** risk 88.1

pressures: homelessness, poverty

- **Preventive tenancy sustainment (pre-Section-21 reach)** — gain 0.85, ~£950
- **Rapid rehousing + TA diversion** — gain 0.80, ~£1,300
- **Rent arrears mediation with landlord** — gain 0.60, ~£700
- **Targeted child-poverty cash / cost-of-living support** — gain 0.90, ~£1,100
- **Debt advice + money guidance (CHW-delivered)** — gain 0.70, ~£650
- **Living-wage / employment access programme** — gain 0.50, ~£1,400

Stop / escalate: Review at week 8; stop expansion if completion <35% or access gap widens in Manchester. Escalate to safeguarding lead if self-harm risk indicated.

Critical **Kingston upon Hull** risk 87.6

pressures: poverty

- **Targeted child-poverty cash / cost-of-living support** — gain 0.90, ~£1,100
- **Debt advice + money guidance (CHW-delivered)** — gain 0.70, ~£650
- **Living-wage / employment access programme** — gain 0.50, ~£1,400

Stop / escalate: Review at week 8; stop expansion if completion <35% or access gap widens in Kingston upon Hull. Escalate to safeguarding lead if self-harm risk indicated.

6 — Live news early-warning

32 live signals · early-warning { 'poverty': 'ELEVATED', 'homelessness': 'ELEVATED', 'nhs': 'ELEVATED', 'mental': 'ELEVATED', 'isolation': 'baseline' }

7 — Second-order: MiroFish agent reaction

(1) Community pressure will concentrate on inner-city areas with vulnerable families facing reduced benefits. (2) Three concrete interventions are: - Targeted assistance such as utility subsidies and emergency food supplies for vulnerable families. - Prevention spending on services like job training programs and affordable housing initiatives. - Expanding public health campaigns to include mental health support resources.

Mental health and ageing are the primary modelled domains per the CIC architecture; the 4th Space is the flagship loneliness-combat intervention. Priors are transparent and illustrative; trend projection is planning-only (one slice + live news momentum), not a calibrated forecast. Replace with local evaluation data before any spend. No individual-level inference. Aggregate place-level only.
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